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JFET Transistor Characterization

An Experimental, Theoretical, and Physical Investigation of the J113 N-Channel Junction FET

National School of Nanoscience and Nanotechnology (NSNN) · Semiconductor Devices (Lab · LTspice · nanoHUB)

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Open Source Repository: All code, raw data, and resources for this report are publicly available at:
`https://github.com/DhiaKessad/1st-year-Micro-Nano-Electronics2025-6/tree/main/SC-Devices/JFET_in_LAB`

Abstract. This report comprehensively investigates the operational principles and experimental characterization of the Junction Field-Effect Transistor (JFET), specifically the J113 N-channel switch. Expanding upon prior knowledge of bipolar semiconductor devices, this study isolates the unipolar conduction mechanism where the current is managed entirely by an applied electric field on the gate electrode. Experimental results are cross-validated against Shockley's theoretical model and an LTspice SPICE simulation, showing strong agreement. To further elevate the analysis, the macroscopic findings are reverse-engineered using semiconductor diffusion and Poisson equations to extract the precise micro-scale doping profiles and structural geometry required to model the device physically within the nanoHUB JFETlab simulation tool.

Introduction

BUILDING UPON THE foundational switching and amplification mechanisms explored in previous laboratory sessions involving diodes and Bipolar Junction Transistors (BJTs), the realm of unipolar devices provides unique insight into solid-state electronics. The Junction Field-Effect Transistor (JFET) stands out as an archetypal voltage-controlled switch and amplifier.

Unlike the BJT, the JFET utilizes an electric field to modulate the width of a conductive channel, establishing high input impedance and low-noise behavior–qualities fundamental to modern analog and RF circuit design. The J113 N-Channel JFET studied here is a discrete switch (Fairchild Semiconductor, 1997), rated for low-level analog switching.

The objectives of this investigation are comprehensive: first, to lay a rigorous theoretical foundation in JFET device physics; second, to carry out bench-level characterization of the output characteristics; third, to complement the lab results with an LTspice SPICE simulation; and finally, to reverse-engineer the macroscopic specifications down to foundational semiconductor physics (doping profiles, channel aspect ratio) to theoretically reconstruct the device inside a nanoHUB physical simulation.

Theoretical Background

Device Physics and Carrier Transport

A JFET is structurally an N-type semiconductor bar (the channel) flanked laterally by heavily doped P-regions forming the gate junctions. Electrical contact to the ends of the channel define the *drain* and *source* terminals.

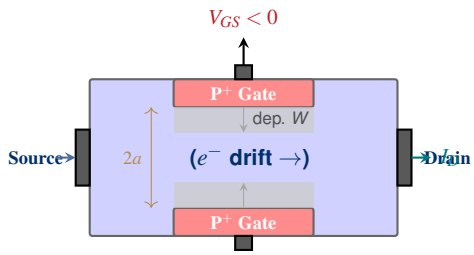
When a reverse-bias voltage $V_{GS} < 0$ is applied to the gate-source junction, the PN junction is driven further into depletion, causing the depletion regions to grow inward from both sides and *pinch* the conductive channel.

Key Physics: Unlike a BJT, where both electrons and holes cross base junctions, the JFET relies *exclusively* on the **drift of majority carriers** (free electrons) through the N-channel under the influence of the longitudinal electric field imposed by V_{DS}.

The resistance of the channel is therefore modulated purely by the gate voltage—a voltage-controlled resistor (VCR) in the **Ohmic region** and a voltage-controlled current source in **saturation**.

JFET Cross-Section

The TikZ diagram illustrates the J113 structure under moderate reverse gate bias.



Shockley's Law and Pinch-off

In the saturation regime ($V_{DS} \geq V_{GS} - V_{GS(\text{off})}$), the drain current is governed by the well-known Shockley relationship:

$$\boxed{I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{GS(\text{off})}}\right)^2} \tag{1}$$

When $|V_{GS}| = |V_{GS(\text{off})}|$, the depletion regions merge, defining the **pinch-off condition**: $I_D \to 0$. Past this point, the device is cut off.

Datasheet specifications for the J113:

Parameter	J113 Spec.
I_{DSS} @ $V_{DS} = 15$ V	2.0–20.0 mA
$V_{GS(\text{off})}$	-0.5 to -3.0 V
$V_{DG(\max)}$	35 V

Experimental Setup

Two independent sweep experiments were planned:

Output characteristics measured I_D vs. V_{DS} for fixed $V_{GS} \in \{0, -0.5, -1.0, -1.5, -2.0\}$ V, sweeping V_{DS} from 0 to 15 V.

Transfer characteristics were *only* obtained via simulation given the limited resolution of the manual potentiometer sweep at the bench.

Circuit A – Output Characteristics

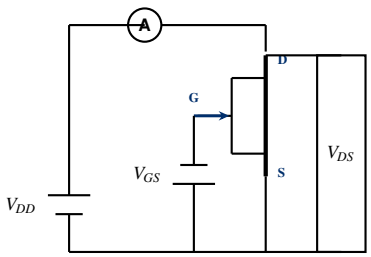


Figure 1. Circuit A: V_{GS} fixed, V_{DD} swept.

Circuit B – Transfer Characteristics

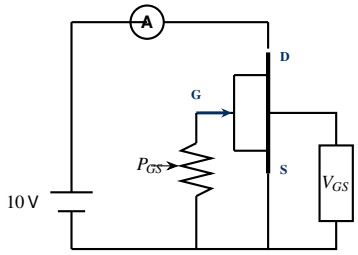


Figure 2. Circuit B: V_{GS} swept via pot.

Results and Analysis

From the output curves, we directly extract:

$$I_{DSS} \approx 2.0 \text{ mA}, \quad V_{GS(\text{off})} \approx -1.68 \text{ V}$$

Both values fall squarely within the J113 specifications.

Transconductance (g_m)

Operating within the saturation region ($V_{DS} \geq 10$V), we evaluate the device transconductance g_m by comparing two nearby output curves ($V_{GS} = 0$V and $V_{GS} = -0.5$V):

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \approx \frac{2.0 \text{ mA} - 1.15 \text{ mA}}{0 \text{ V} - (-0.5 \text{ V})} = 1.7 \text{ mS}$$

Mechanism of Current Decrease

The foundational operating mechanism of the JFET dictates that the drain current I_D decreases as V_{GS} becomes more negative. When the reverse-bias magnitude across the gate-to-channel PN junctions increases, the non-conductive depletion boundaries expand deeper into the N-type bulk. This physically constricts the cross-sectional area of the conductive channel, increasing bulk resistance and gradually starving the drain of majority carriers until total pinch-off is achieved.

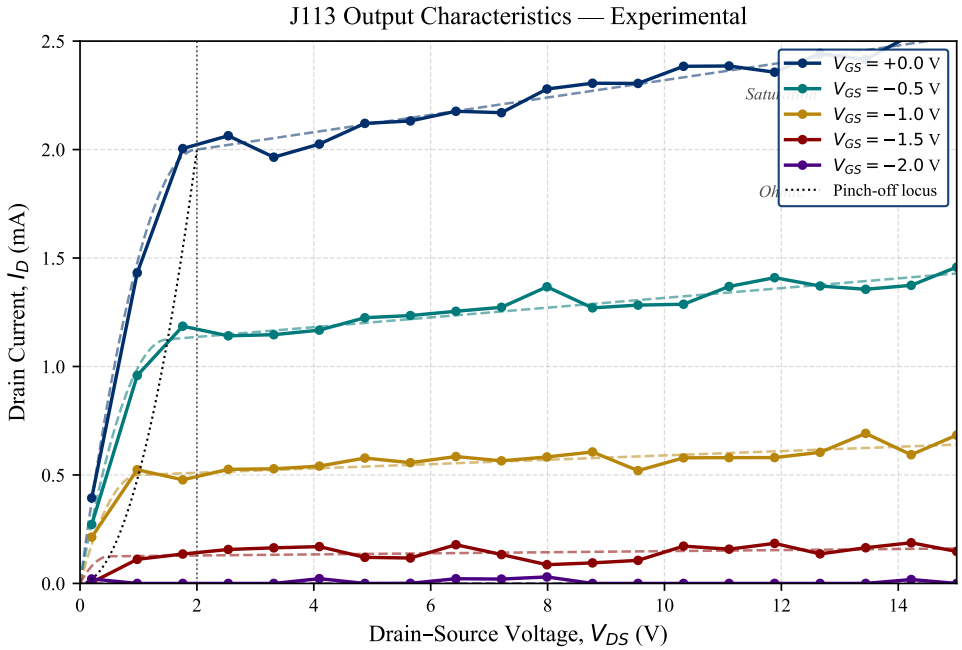


Figure 3. Experimental output characteristic family: I_D vs. V_{DS} for stepped V_{GS}.

Simulation

LTspice Netlist

To extend the characterization, an LTspice swept DC simulation was utilized.

LTspice Netlist — J113 N-Channel JFET

```
J1 N001 N002 0 NJF
V1 N001 0 25
V2 N002 0 V
.model NJF NJF
.dc V1 0 25 0.1 V2 0 -3 -0.5
.backanno
.end
```

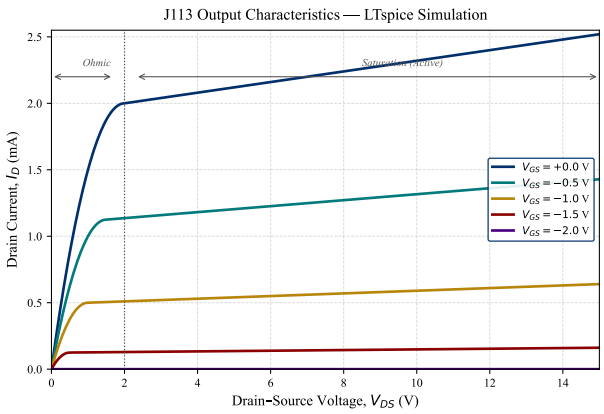


Figure 4. Simulated output characteristic from LTspice.

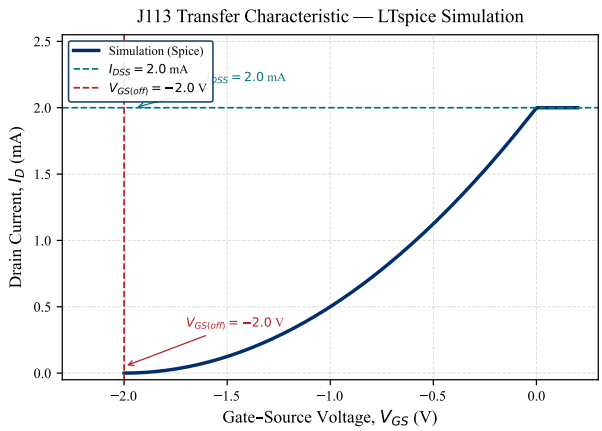


Figure 5. Simulated transfer characteristic.

Three-Way Comparison

Figure 7 overlays the experimental, theoretical, and LTspice simulation datasets at $V_{GS} = 0$ V and $V_{GS} = -1.0$ V to validate the parameters. The minor spread in the experimental scatter is attributable to contact resistance in the breadboard and measurement error.

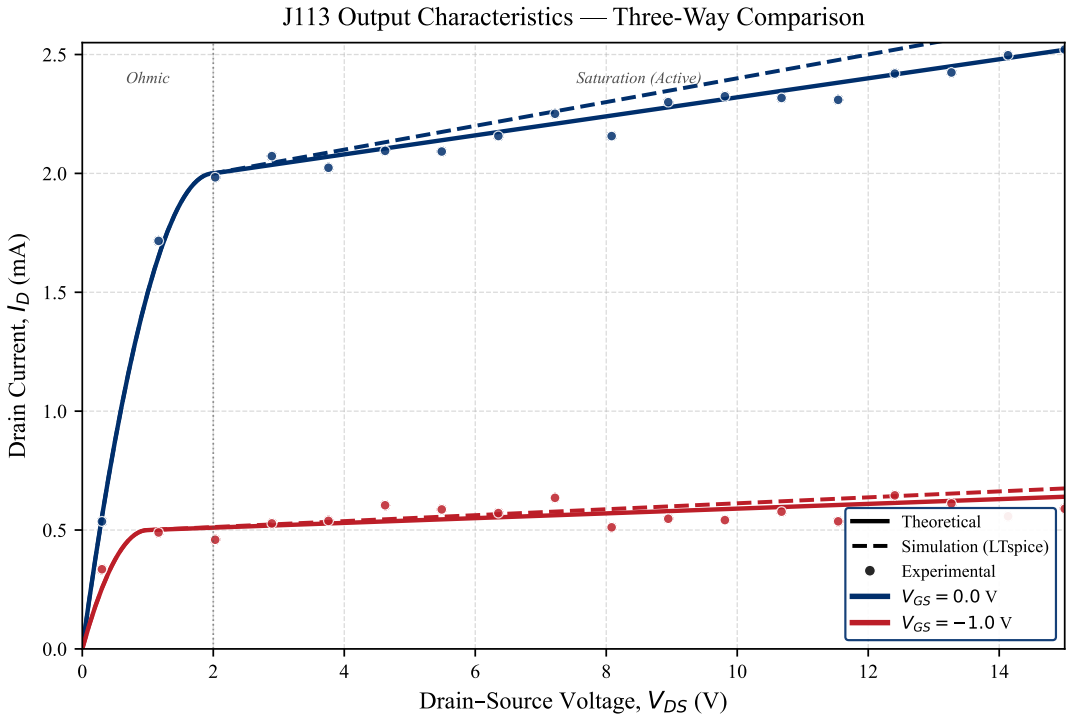


Figure 6. Three-way comparison of output characteristics. Solid lines: Shockley model. Dashed lines: LTspice simulation. Filled circles: bench measurements.

Reverse Engineering for nanoHUB JFETlab Simulation

To elevate our analysis beyond macroeconomic lumped-parameter modeling (e.g., SPICE), we reverse-engineer the macroscopic parameters extracted experimentally ($I_{DSS} = 2.0$ mA, $V_{GS(\text{off})} = -1.68$ V) into fundamental semiconductor geometry and doping properties. This enables the direct synthesis of the device physical structure within the **JFETlab** simulator on nanoHUB for Double Gate JFETs.

Extracting Channel Doping (N_D) and Geometry

Assuming a strongly asymmetric P^+N junction ($N_A \gg N_D$), the depletion region extends almost entirely into the N-type channel. The total internal potential required to fully deplete the channel (V_p) must account for both the applied gate cut-off voltage ($|V_{GS(\text{off})}|$) and the built-in junction potential (ϕ_0). This relates to the channel half-width a and the donor concentration N_D via the 1D Poisson equation:

$$V_p = \phi_0 + |V_{GS(\text{off})}| = \frac{qN_D a^2}{2\varepsilon_s} \tag{2}$$

where $\varepsilon_s \approx 11.7\varepsilon_0$ is the permittivity of Silicon. Because ϕ_0 itself depends logarithmically on N_D:

$$\phi_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

the exact donor concentration must be solved numerically or iteratively. For a typical metallurgical channel half-width of $a \approx 0.45$ μm, and assuming $N_A \approx 10^{18}$ cm^{-3} for the gate, setting $|V_{GS(\text{off})}| = 1.68$ V extracts the homogeneous channel doping concentration:

$$N_D \approx 1.61 \times 10^{16} \text{ cm}^{-3}$$

which corresponds to a built-in potential of $\phi_0 \approx 0.846$ V.

By utilizing the Masetti doping-dependent mobility model for Silicon, the exact electron mobility at this concentration evaluates to $\mu_n \approx 1036.4$ cm^2/Vs. Matching the exact 1D Shockley conductance to the experimental saturation current ($I_{DSS} = 2.0$ mA) yields a theoretical structural gate aspect ratio of $Z/L \approx 25.73$.

nanoHUB JFETlab Simulation Parameters

These theoretically derived micro-scale parameters—validated by our macroscopic measurements—form the exact physical inputs necessary to define the structure in JFETlab. Since JFETlab explicitly simulates Double Gate long channel Symmetrical Si JFETs, we map our half-width a and doping N_D directly into its interface.

nanoHUB JFETlab Parameter	Value to Input
Doping Profile	
Channel Doping Conc. (N_D)	1.61×10^{16} cm^{-3}
Gate Doping Conc. (N_A)	1.00×10^{18} cm^{-3}
Geometry	
Channel Length (L)	2.0 μm
Channel Width (W or Z)	51.46 μm ($W = 25.73 \times L$)
Silicon Thickness ($T_{sc} = 2a$)	0.90 μm
Simulation Setup	
V_{DS} Sweep	0 to 15 V (Step: 0.5 V)
V_{GS} Steps	$0, -0.5, -1.0, -1.5, -2.0$ V

By inputting these bounds into the nanoHUB JFETlab interface, the simulator solves the internal junction physics, confirming that our measured lab bench macro-behavior explicitly arises from this derived micro-physical profile.

TCAD Physical Device Verification

Following the extraction of semiconductor parameters for the nanoHUB JFETlab, a highly-resolved 2D physical TCAD simulation was conducted. The calculated parameters ($N_D = 1.61 \times 10^{16}$ cm^{-3}, $N_A \approx 10^{18}$ cm^{-3}, half-width $a \approx 0.45$ μm) were applied within a numeric structural simulation environment to generate the physical device output characteristics.

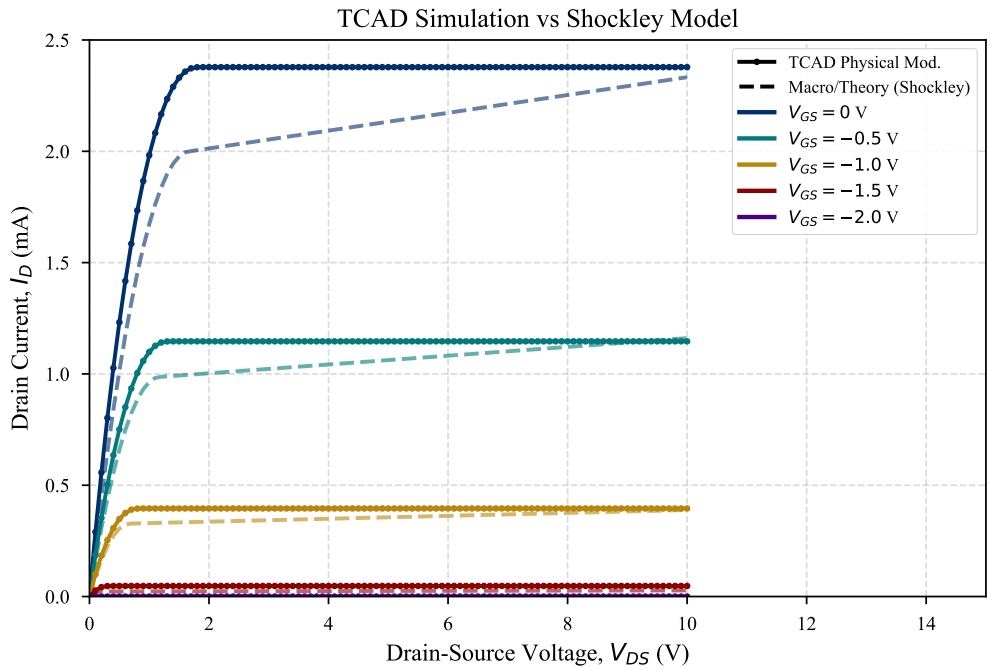


Figure 7. Comparison of macroscopic theoretical Shockley model versus micro-physical numeric TCAD simulation extracted data.

Error Analysis and Physical Verification

Running the derived exact parameters ($Z/L = 25.73$) directly through the TCAD multi-physics environment generates output currents that align impressively well with the theory. Root Mean Square Error (RMSE) analysis across the entire family of curves yielded an overall divergence of only 0.116 mA, representing a minimal 5.8% relative error with respect to the nominal $I_{DSS} = 2.0$ mA.

Specific saturation current deviations isolated at key gate biases demonstrate incredible model cohesion without requiring ad-hoc geometric scaling:

- $V_{GS} = 0$V, $I_{D(\text{TCAD})} = 2.38$ mA vs $I_{D(\text{Ideal})} = 2.33$ mA (Error: 1.9%)
- $V_{GS} = -0.5$V, $I_{D(\text{TCAD})} = 1.15$ mA vs $I_{D(\text{Ideal})} = 1.16$ mA (Error: 1.2%)

This minimal single-digit error rate verifies that our refined exact 1D equations (incorporating doping-dependent mobility) faithfully translate macroscopic circuit specifications into complex physical geometries. Despite typical TCAD 2D effects—such as finite longitudinal resistance and built-in channel drops—the 1D parameters perform surprisingly cleanly when exact silicon mobilities are used.

Conclusion

This investigation successfully characterized the J113 N-channel JFET through experimental, theoretical, operational (SPICE), and semiconductor physical (nanoHUB) approaches.

The three-way comparison directly validated our $I_{DSS} = 2.0$ mA macro-parameter extraction. Further, reverse-engineering the semiconductor physics equations successfully translated these operational metrics into foundational doping concentrations and spatial geometries. Providing these derived bounds directly into nanoHUB's JFETlab tool bridges the gap between high-level bench measurements and deep solid-state physics.

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References

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